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DATE:19.03.2026

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ASSIGNMENT -9

Library Management System using React, NodeJS and MySQL

AIM

To design and develop a full-stack **Library Management System** using React for the frontend, Node.js with Express for the backend, and MySQL as the database, to perform CRUD (Create, Read, Update, Delete) operations on book records with proper routing, user interaction, and real-time data updates.

DESCRIPTION

useState:

Used to store and update component state

syntax: `const [data, setData] = useState(initialValue);`

useEffect:

Used to run code when component loads or updates

syntax: `useEffect(() => { /* code */ }, []);`

useParams:

Used to get parameters from URL

syntax: `const { id } = useParams();`

GET (Axios):

Used to fetch data from backend

syntax: `axios.get(url).then(res => setData(res.data));`

POST (Axios):

Used to send new data to backend

syntax: `axios.post(url, data);`

PUT (Axios):

Used to update existing data

syntax: `axios.put(`${url}/id`, data);``

DELETE (Axios):

Used to delete data

syntax: `axios.delete(`${url}/id`);``

Express Route:

Used to define API endpoints

syntax: `app.get("/path", (req, res) => {});`

MySQL Query:

Used to execute SQL commands

syntax: `db.query("SQL QUERY", values, callback);`

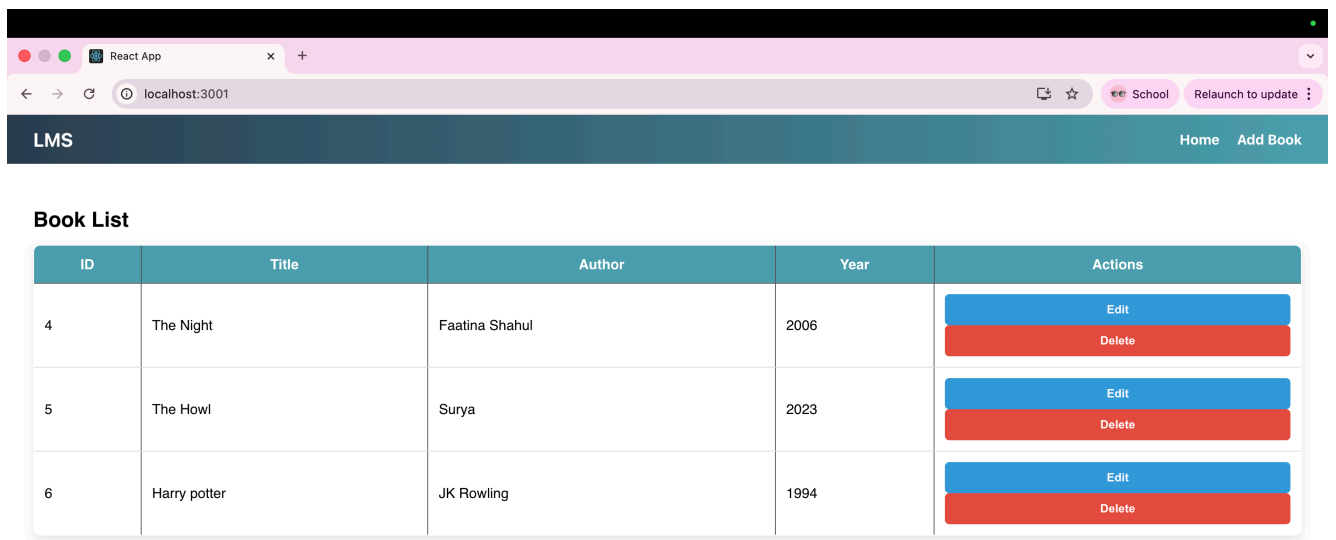
React Router:

Used to navigate between components

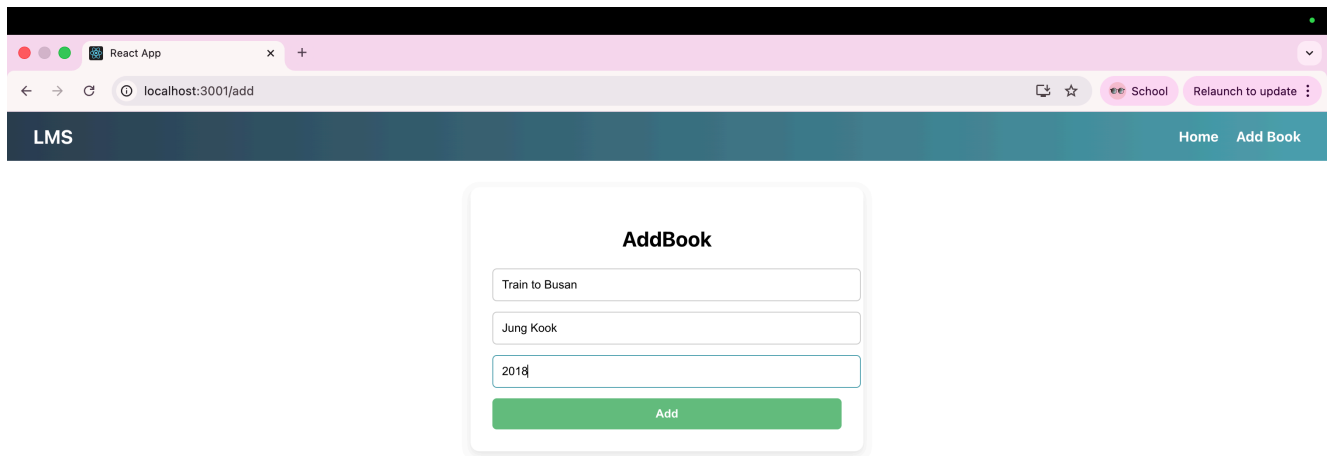
syntax: `<Route path="/path" element={<Component />} />`

OUTPUT

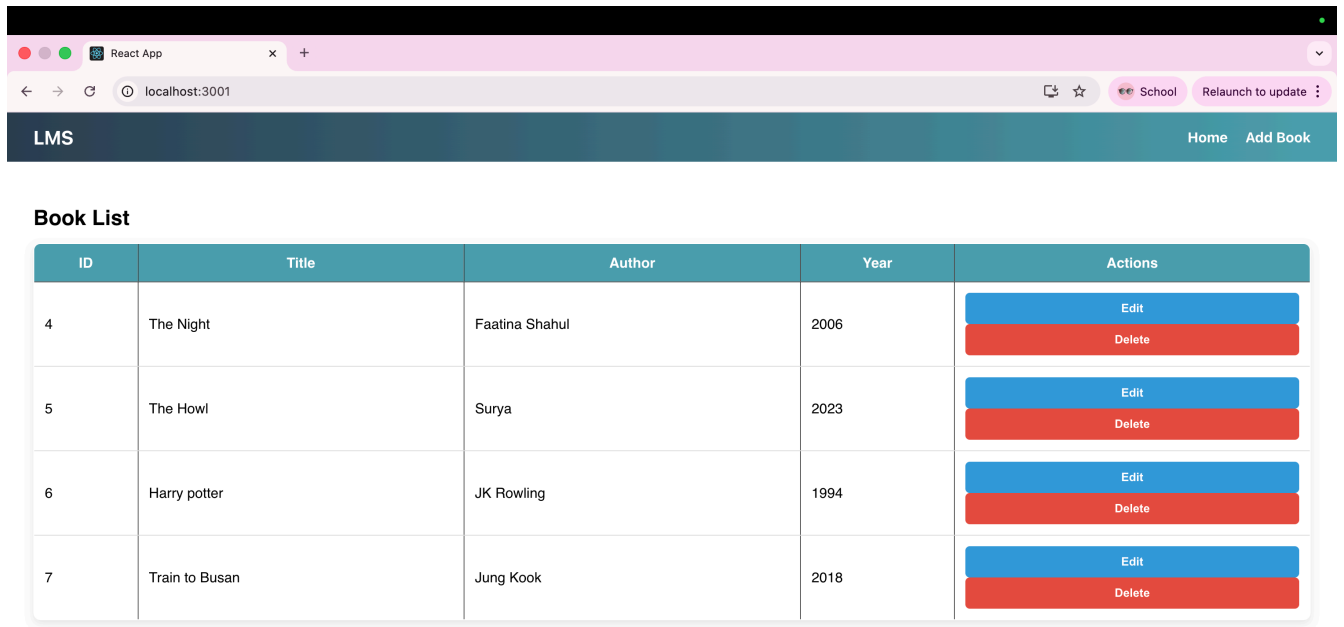
Home page



Adding a Book

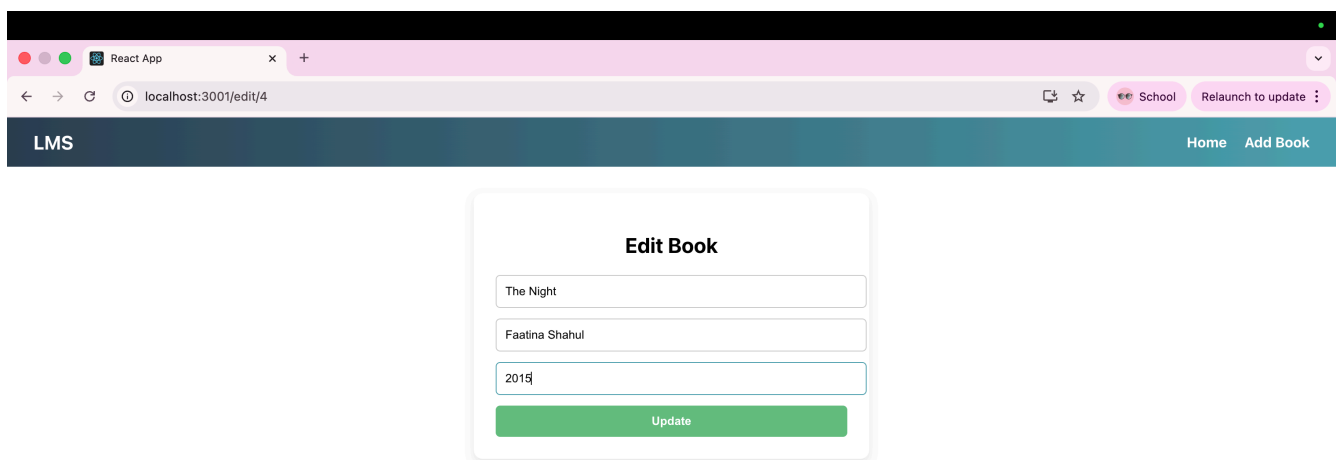


After adding the book “Train to Busan”



ID	Title	Author	Year	Actions
4	The Night	Faatina Shahul	2006	Edit Delete
5	The Howl	Surya	2023	Edit Delete
6	Harry potter	JK Rowling	1994	Edit Delete
7	Train to Busan	Jung Kook	2018	Edit Delete

Editing the book “The Night”



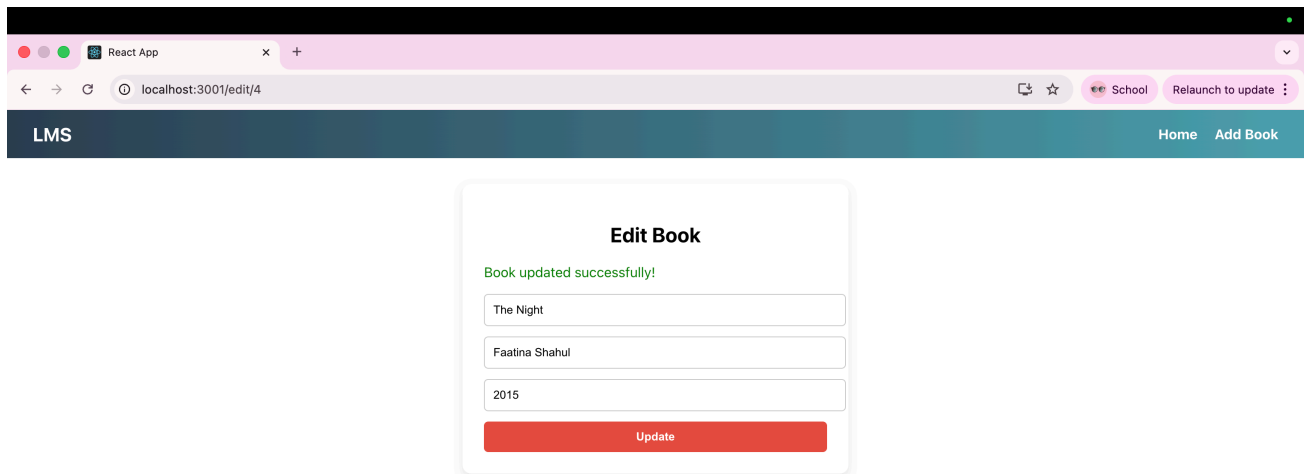
Edit Book

The Night

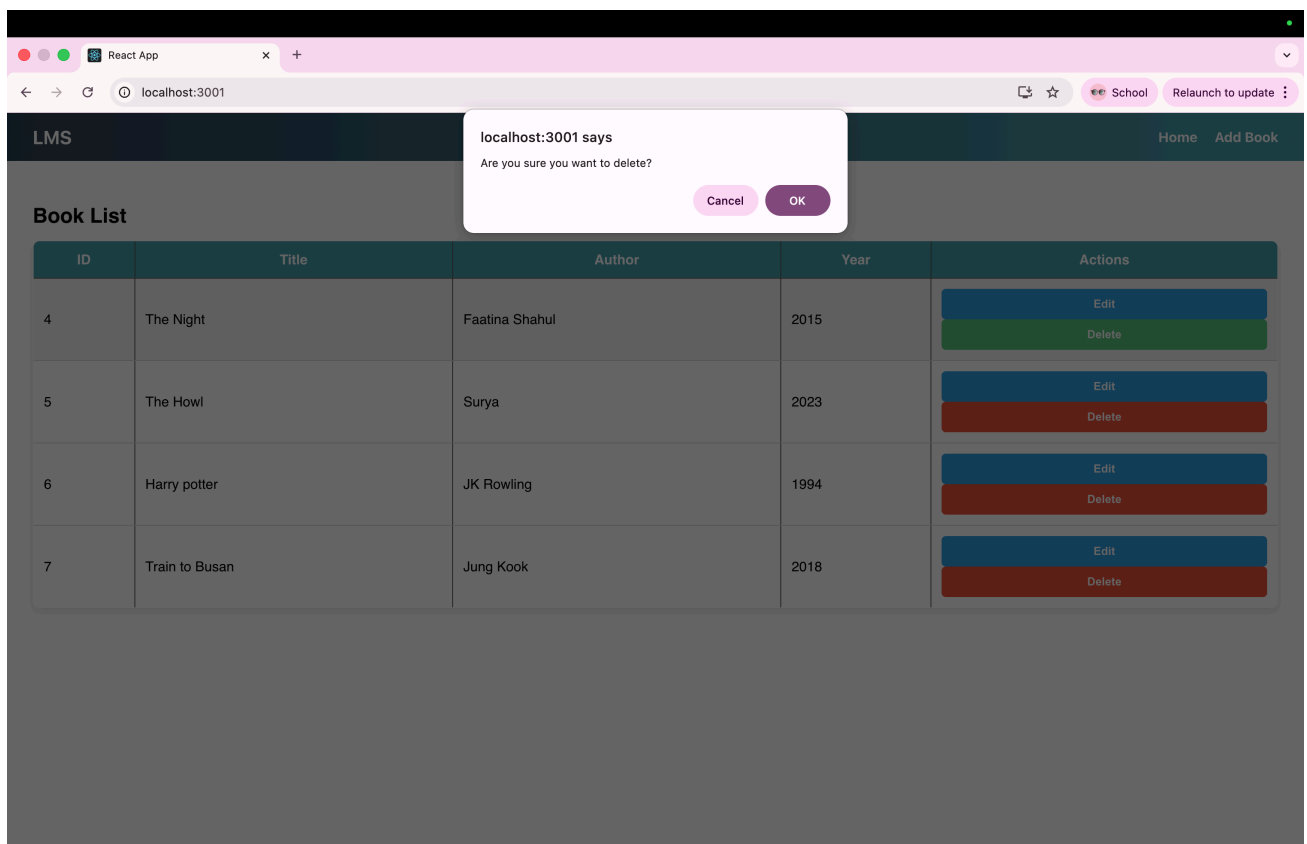
Faatina Shahul

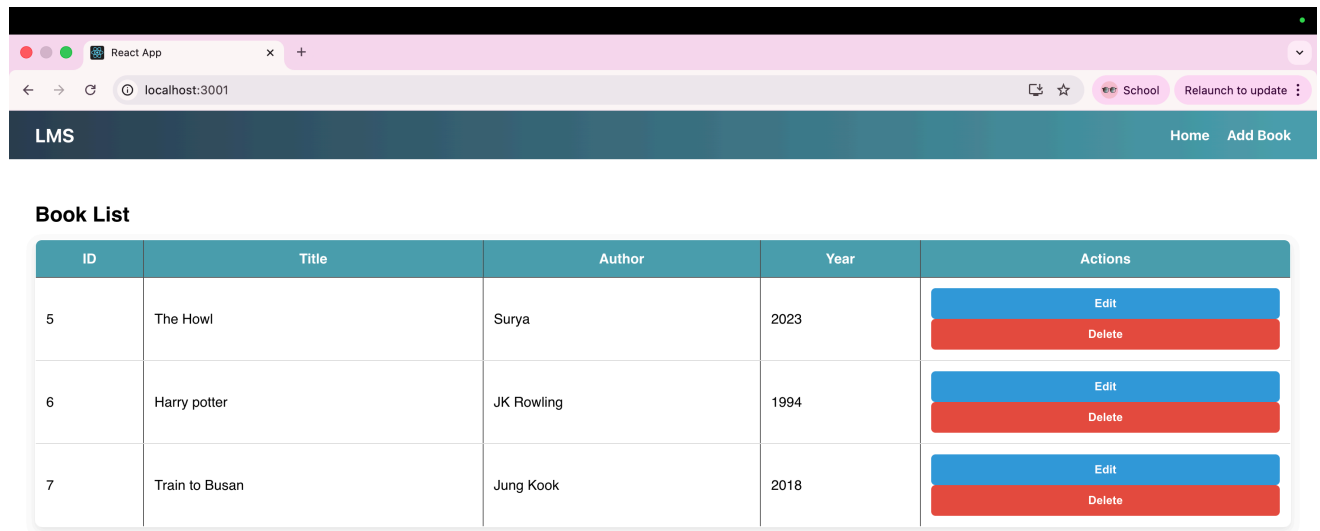
2018

Update



Deleting the book “The Night”





The screenshot shows a web browser window with the title 'React App' and the address 'localhost:3001'. The application displays a 'Book List' table with the following data:

ID	Title	Author	Year	Actions
5	The Howl	Surya	2023	Edit Delete
6	Harry potter	JK Rowling	1994	Edit Delete
7	Train to Busan	Jung Kook	2018	Edit Delete

LEARNING OUTCOME

- Developed a full-stack application integrating React, Node.js, and MySQL for managing library records.
- Implemented CRUD operations using RESTful APIs to add, view, update, and delete book details.
- Designed and navigated multiple React components using React Router for smooth user interaction.
- Connected the frontend with backend services using Axios to handle asynchronous data exchange.
- Ensured dynamic UI updates and basic error handling to reflect changes immediately after operations.